Name:_____ ID:_____

CSE 474 Computer Networks

Fall 2017 Final Exam

04.01.2018 Thursday, Duration: 100 minutes

Q1	Q2	Q3	SUM
/25	/50	/35	/100

Q-I: (25 pts) Transport Layer – Please answer the following questions:

I-1. (8 pts) Remember that timeout value in TCP is calculated as Estimated RTT + 4*DevRTT. Suppose that, at time *t*, Estimated RTT value is 100 ms and the next sample RTT measurements fluctuates like 105 ms, 95 ms, 105 ms, 95 ms, 105 ms, 95 ms at times t+1, t+2, t+3, t+4, t+5 and t+6.

At time t+7, which of the following are true? Justify your answer.

(a – 3 pts) Estimated RTT is more than 100. (Although you may not need, you can assume that α=0,125.)

(b – 5 pts) DevRTT value is more than 5. (Although you may not need, you can assume that β=0,25.)

I-2. (10 pts) In a TCP scenario, host A sends five packets with SEQ=72, SEQ=80, SEQ=88, SEQ=96 and SEQ=104 to host B. Host B sends back four packets all with ACK=80. Answer the following questions:

(a – 2 pts) What is the length of first four packets sent by host A?

(b – 4 pts) Can you tell whether a packet or ACK is lost or not? If yes, which one?

(c – 4 pts) After receiving four packets with ACK=80, what would be the SEQ value of the next packet sent by host A? Why?

I-3. (3 pts) What is the purpose of receive window (rwnd) field in the TCP header?

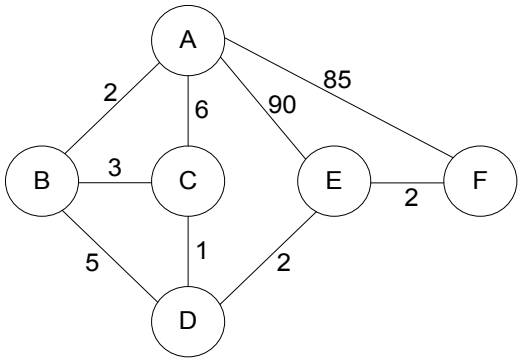
I-4. (4 pts) Is there any fairness problem when there are both TCP and UDP traffic on a link? Briefly describe.

Name:_____ ID:_____

Q-II: (50 pts) Network Layer – Please answer the following questions:

II-1. (21 pts) Consider the following graph:

(a – 9 pts) Show the operation of Dijkstra's (Link State) algorithm for computing the least cost path from C to all destinations. How does C set its forwarding table entries for all destinations if the traditional destination-based forwarding is used?

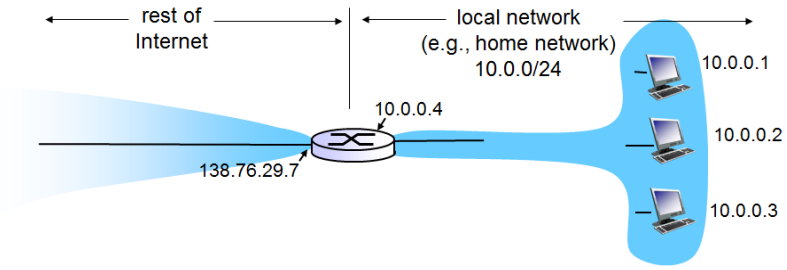


(b - 4 pts) Suppose that we use Software Defined Networking (SDN) approach (Openflow) and we want all the packets send from A to D sent over path A-C-D. How should we set match and action fields in nodes A and C?

(c – 3 pts) Suppose that Distance Vector Routing is used and all the distance vectors are converged. What happens when link (E,A) is broken?

(d – 5 pts) Removal of which link causes count-to-infinity problem? How? Show your work.

II-2. (13 pts) Consider the following NAT scenario. Answer the following questions.



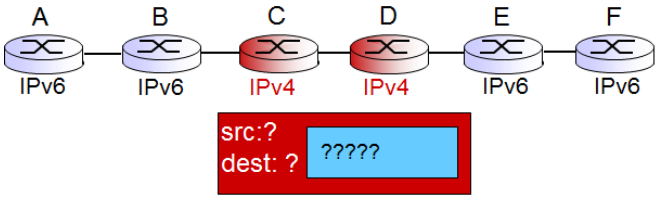
(a – 4 pts) Suppose that a process running at host with IP address 10.0.0.2 and port 8973 sends a packet to outside server with IP address 192.196.24.34. Will the IP header and TCP header of this packet change? If yes how? If no why?

Name:_____ ID:_____

(b – 4 pts) Can you implement NAT using software defined networking approach? How?

(c – 5 pts) Suppose that Ali is in university and wants to perform a remote access to his home computer running at 10.0.0.1 using TeamViewer from another city. What is the problem? Give a possible solution. (You don't need to know the exact solution of Teamviewer, but give a reasonable answer).

II-3. (5 pts) Consider the following network where some of the routers are IPv6 enabled, but some of them (C and D) are not. Suppose that A sends a message encapsulated in an IPv6 datagram to F. For this message, describe the IP source and IP destination addresses and content of the packet sent from C to D.



II-4. (3 pts) In hierarchical routing, what is a gateway router?

II-5. (8 pts) Suppose that a link between Router 1 and Router 2 has a maximum transfer unit (MTU) of 5000 bytes. Answer the following questions:
(a – 3 pts) If an IPv4 datagram of size 10000 bytes arrive to the link from Router 1 to Router 2, this datagram has to be fragmented into smaller datagrams. How many datagrams will be sent to Router 2?

(b – 2 pts) In which node are these datagrams reassembled?

(c – 3 pts) If these datagram was an IPv6 datagram, what would be the action taken by Router 1?

Q-III. (35 pts) Link Layer and LAN – Please answer the following questions:

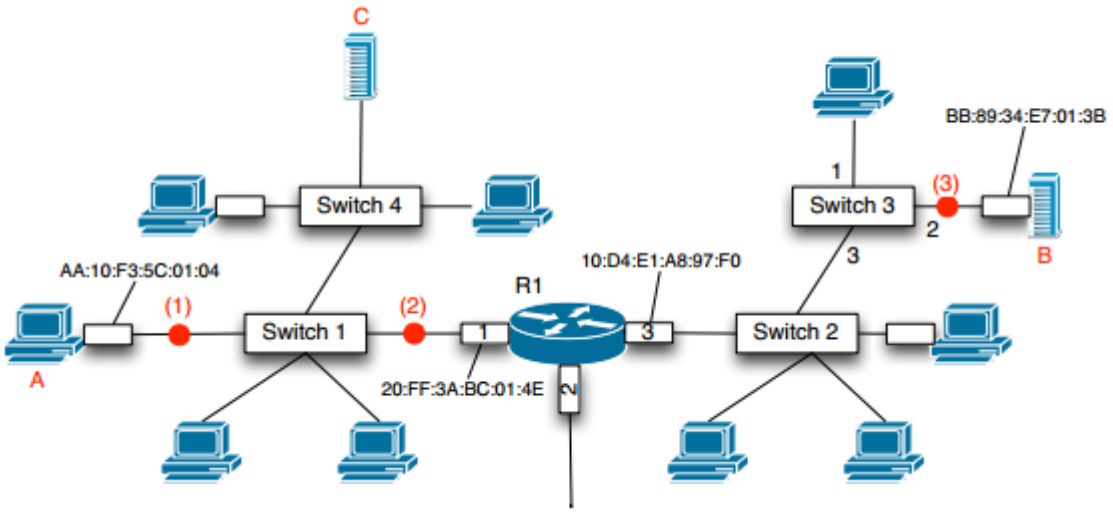
III-1. (4 pts) In CSMA/CD, briefly describe the relation between propagation delay and collision probability (give also the reason).

III-2. (4 pts) For wide area cellular networks (such as 3G or LTE), channel partitioning MAC protocols are more appropriate than random access MAC protocols. What could be a reason?

III-3. (5 pts) In Cyclic Redundancy Check (CRC), suppose that 5-bit generator G = 10001, and data D = 1100110011. Find the additional bits R that the sender will append to D.

Name:_____ ID:_____

III-4. (22 pts)



Consider the LAN scenario above. Suppose all switches are learning switches. All the ARP tables and switch tables are initially empty. Answer each question below *briefly*, e.g., *in a sentence or two at most.*

a) (3 pts) Suppose that B wants to send a packet to A. Consider an ARP request sent by node B. Whose IP-to MAC address translation is being queried?

b) (3 pts) What is the destination MAC address on the frame containing the ARP request?

c) (4 pts) After B receives the ARP reply, what is contained in switch 3 switching table?

d) (6 pts) Suppose that A has an IP address of 193.140.1.3. Subnet mask for both subnets is 22.
(i) Is 193.140.2.230 a possible IP address for C? Why?

(ii) Is 193.140.3.15 a possible IP address for B? Why?

e) (3 pts) Now consider the frame containing the B-to-A IP datagram. What are the MAC source and destination address of this frame and the IP source and destination addresses in the encapsulated IP datagram at points (1), (2), and (3)? (For IP addresses, just say IP of A, IP of B, etc)

f) (3 pts) Consider the datagram being sent from A to C; neither A nor C have sent any frames or datagrams in the network before. How many of hosts in the network receive the frame containing the datagram sent by A? Explain your answer briefly.